

Dynamic Moral Trajectory for Emotional Manipulation Detection using Moral Foundations Theory

1st Joshua Joyson Kustiadi
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
joshua.kustiadi@binus.ac.id

2nd Kelly Roselin Wijaya
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
kelly.wijaya002@binus.ac.id

3rd Raynaldo Winson Amadeus Kusuma
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
raynaldo.kusuma@binus.ac.id

4th Kristien Margi
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
kristien.margi@binus.ac.id

5th Ricky Reynardo Siswanto
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
ricky.siswanto@binus.ac.id

Abstract— This research proposes the Moral Hijacking Detector, an AI-powered Natural Language Processing (NLP) framework based on Moral Foundations Theory (MFT). Unlike prior studies that rely on static moral or sentiment features, this work introduces a Dynamic Moral Trajectory Engineering approach that models how moral framing evolves throughout a text using features such as switching rate, escalation dynamics, instability index, moral entropy, and vice-dominance ratio. The framework further integrates VADER-based sentiment volatility and stylistic linguistic features to create a multi-dimensional representation of manipulative narratives. The framework is evaluated using a combined dataset consisting of a propaganda dataset containing emotionally manipulated statements and a Tweet Emotions dataset, where emotionally charged categories such as fear, sadness, joy, and love are labeled as manipulative signals while neutral content is labeled as non-manipulative. We tried multiple configurations using StratifiedKFold cross-validation. The best-performing Stacking Ensemble is achieved by combining XGBoost base models with Random Forest and LightGBM as individual base models, using Logistic Regression as a meta-learner giving the mean AUC-ROC of 0.823 and Macro-F1 of 0.736 respectively. Results indicate identifiable signatures of escalating moral intensity, volatile moral transitions and vice-dominant framing that characterize manipulative stories meaning-directed activation as a parsimonious approach to emotion-based persuasion validates dynamic trajectory analysis as an explanatory mechanism for manipulative narrative detection.

Keywords— *Emotional Manipulation Detection, Dynamic Moral Trajectory, Moral Foundations Theory, Propaganda Detection, Natural Language Processing, Explainable AI, Computational Social Science.*

I. INTRODUCTION

Today the increased reliance upon our connected society can leave individuals susceptible to emotive manipulation through the digital information [7]. This emotional manipulation, driven from financial gain [2] or ideological perspectives [11] influences public mood and can create emotional reactions of guilt [6] or anxiety [4], overcoming individual judgement.

Though recently some researchers have focused on this issue, the detection of emotive manipulation is still very hard, since standard fact-checking approaches are too simplistic to reflect tone, intent and moral framing [3], whereas existing models represent emotion and morality with fixed ways,

failing to reflect how emotion and morality changes throughout a document, and often ignore diverse moral views among individuals that affect the outcome of the detection [5][10].

This study aims to develop a solution using an AI powered Natural Language Processing (NLP) tool called “Moral Hijacking Detector” based on the theory of Moral Foundations Theory (MFT) [10]. It introduces dynamic moral trajectory engineering to measure the change of moral discourse by considering features like switching rate, escalation dynamics, instability index, moral entropy and vice-dominance ratio together with other VADER sentiment features and stylometric features. It then used a stacking ensemble of ML models (XGBoost, Random Forest, and LightGBM with Logistic Regression) [8][9][14][15] on the propaganda and Tweet Emotions dataset.

This paper contributes towards not just emotion manipulation detection with static data, but rather extends towards a dynamic yet interpretable framework which aims to aid applications for explainable AI and for building digital literacy.

II. RELATED WORKS

Previous studies highlight the capacity of different moral framings, such as instilling a feeling of guilt or evoking moral outrage, to powerfully shape decisions and social behavior [1], [2], [4], [6], [7]. Nonetheless, such studies are mostly observational, based on self-reports, and there is a dearth of scaled approaches for detection of manipulation in texts.

In the Computational perspective, MFT has been used for text analysis to reveal patterns in discourse of persuasion [3], while sentiment analysis and other AI based methodologies can reveal the emotions underlying linguistic expressions [16][17]. Machine learning models, such as XGBoost [8][9] and LightGBM [12][13] perform very well on the problem of high-dimensional and unbalanced datasets. However most of the approaches available use static features and do not deeply capture the evolution of psychological and narrative states.

All previous works are fragmented from psychological theory to computational text analysis and machine learning, lacking an approach that can holistically incorporate and dynamically track the moral trajectory of a narrative. In view of these deficiencies, this research aims to design a Dynamic

Moral Trajectory Engineering method in the Moral Hijacking Detector frame integrating MFT-based dynamic features (e.g. moral entropy, switching, escalation), volatility of sentiments and ensemble learning.

III. METHODOLOGY

We propose a multi-stage text analysis pipeline for emotional manipulation detection.

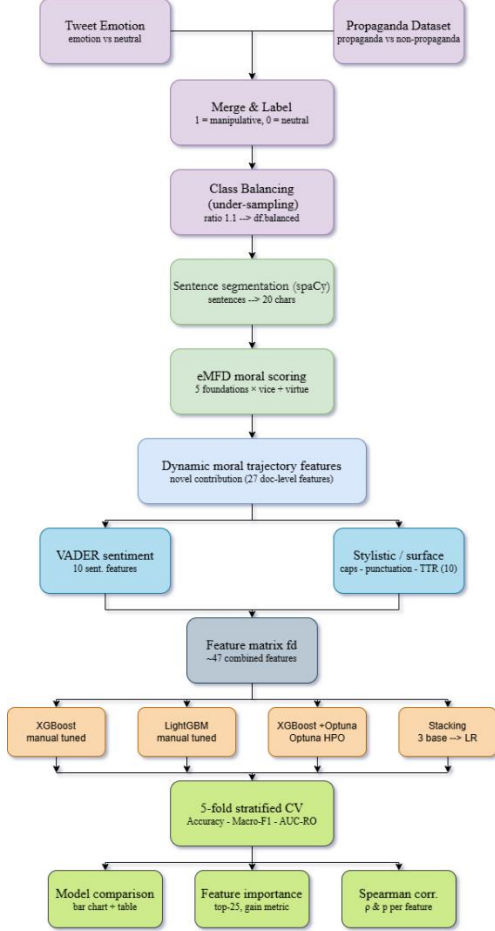


Fig. 1. Workflow pipeline

A. Dataset Collection and Description

This research uses two datasets; the Tweet Emotions dataset and the Propaganda dataset, which are chosen due to its representation of both emotion expression and rhetorical manipulation units in tweets with around 30,376 data for training samples.

TABLE I. DATASET OVERVIEW

Attribute	Tweet Emotions Dataset	Propaganda Dataset
Source / Platform	Twitter / Kaggle	SemEval / NLP4IF
Original Labels	neutral, empty, worry, happiness, sadness, love	propaganda, non-propaganda
Mapped Label = 0 (Neutral)	neutral, empty	non-propaganda

Mapped Label = 1 (Manipulative)	worry, happiness, sadness, love	propaganda
Analytical Role	Emotional manipulation signal	Rhetorical propaganda signal

As set out by Moral Foundations Theory (MFT) [10], emotionally driven states associated with worry, sadness, happiness and love are established psychological mechanisms that phenomenologically activate moral foundations. Because they have the ability to cognitively manipulate rational judgement through overt induction of emotional states, they will also inhibit critical thinking abilities [1][4][6]. Therefore, tweets expressing these emotions will be categorized in the manipulative class (label = 1), while neutral and empty tweets (with no overt emotional loading) will be placed within the non-manipulative class (label = 0). The original SemEval/NLP4IF [10] annotating process of the Propaganda dataset is based on this classification process, where propaganda/expression of propaganda is defined as a type of content that uses rhetoric/emotional appeals to induce authors' opinions on readers' opinion and behaviour, and thus is inherently manipulative [3].

B. Data Balancing and Sentence-Level Text Processing

The data is first undersampled against the majority class to counteract class imbalance [9]. The data is sorted into majority (manipulative) and minority (non-manipulative) groups. Then the size of the majority group is resampled randomly (without replacement) to an approximately 110% size of the minority group size,

$$N_{majority}^{sampled} = [1.1 \times N_{minority}]$$

which helps avoid information loss by reducing minority data by strict 1:1 and but keeps both groups sizes near equal so that there are still plenty of examples of majority texts to learn from.

The new smaller groups are then merged, shuffled and reindexed. Next, sentence based processing of the texts is performed using spaCy [12]. Each document is divided into separate sentences and we keep only those sentences that are more than 20 characters long (to exclude meaningless empty sentences and maintain useful semantic information) and attach document ID and positional index of each sentence.

C. EMFD for Moral Feature Extraction and Aggregation

Moral Feature

Extraction is implemented by the Extended Moral Foundations Dictionary (eMFD) [10]. Based on a bag-of-words approach, sentences are scored on five moral foundations, considering both virtues and vices.

$$V(s) = care_v, care_s, fair_v, fair_s, lov_v, lov_s, auth_v, auth_s, sanc_v, sanc_s$$

To calculate the sentence's score with the bag-of-words approach(score_type='bow') on the probabilistic mapping (prob_map='all') and outputs in both vice-virtue format(out_metrics='vice-virtue'), it generated a 10-dimensional moral score vector for each sentence.

This process generate the continuous moral features for text and gives out the morality of text in the continuous spectrum. Eventually, the moral features are integrated with

document ID and sequence of the sentence, into a structured format, to represent semantic information along with positioning information in text.

D. Dynamic Moral Feature Engineering

Instead of treating morality as stationary document-wide averages, which some previous works do [3][10], we model how sentences contribute to morality iteratively as the document progresses. Manipulative texts are characterized not merely by the presence of moral language, but by deliberate escalation and framing shifts designed to progressively override rational judgment. To capture these intra-document patterns, this study extends traditional moral scoring by introducing dynamic feature engineering based on sentence-level moral trajectories.

Using the previously computed eMFD scores, each sentence s is enriched with derived indicators. Moral intensity is defined as the total moral activation across all foundations:

$$I(s) = \sum_{f \in F} (\text{virtue}_f(s) + \text{vice}_f(s))$$

The vice-virtue ratio per sentence is computed as:

$$WR(s) = \frac{\sum_f \text{vice}_f(s)}{\sum_f \text{virtue}_f(s) + \varepsilon}$$

where ε is a small constant to prevent division by zero. The dominant moral frame $\text{dom}(s)$ is defined as the foundation f with the highest combined virtue and vice score for sentence s .

These sentence-level indicators are then aggregated at the document level to compute higher-order metrics. The switching rate between dominant moral frames is defined as:

$$SR = \frac{\sum_{i=1}^{N-1} 1[\text{dom}(s_i) \neq \text{dom}(s_{i+1})]}{N-1}$$

Moral entropy, measuring the diversity of moral foundation usage across the document, is defined as:

$$ME = - \sum_{f \in F} p_f \log_2(p_f)$$

where p_f is the proportional contribution of foundation f to total moral intensity. The instability index captures intra-document fluctuation in moral intensity:

$$II = \sigma(I(s_1), I(s_2), \dots, I(s_N))$$

In addition, trajectory-based features are introduced to model the evolution of emotional signals across the text. Initial hook density and final escalation are computed as the average moral intensity over the first and last two sentences respectively:

$$IHD = \frac{1}{2} \sum_{i=1}^2 I(s_i), \quad FE = \frac{1}{2} \sum_{i=N-1}^N I(s_i)$$

The escalation delta captures the net directional change in moral intensity from the opening to the closing of the text:

$$ED = FE - IHD$$

The trajectory slope is obtained via ordinary least squares regression over sentence order, formally defined as:

$$\beta = \frac{\sum_{i=1}^N (i - \bar{i})(I(s_i) - \bar{I})}{\sum_{i=1}^N (i - \bar{i})^2}$$

where $\bar{i}$ and $\bar{I}$ denote the mean sentence index and mean moral intensity respectively. Furthermore, per-foundation statistics, mean virtue, mean vice, and intensity standard deviation, are computed for each of the five moral dimensions (care, fairness, loyalty, authority, sanctity), yielding 15 additional features.

E. Sentiments and Stylistic Feature Augmentation

We calculate Sentiment features based on VADER lexicon [16][17] at sentence level and aggregate at document level, which comprise: mean, standard deviation, range, and proportion of negative, positive, neutral, and compound sentiment scores at sentence level. In particular, tone volatility is explicitly operationalized as the standard deviation of compound sentiment scores per sentence, which is a useful indicator for emotion-heavy content.

$$TV = \sigma(c_1, c_2, \dots, c_N) = \sqrt{\frac{1}{N} \sum_{i=1}^N (c_i - \bar{c})^2}$$

where c_i is the compound score for the VADER polarity analysis of sentence i , and $\bar{c}$ is the average compound score among all N sentences in the document.

For style features, we extract the features relevant to rhetorical patterns that are commonly found in manipulated text, such as punctuation density (number of exclamation marks, question marks, ellipsis per 100 tokens), percentage of ALL-CAPS words, type-token ratio (lexical diversity), percentage of negative words, length in terms of tokens, characters and average sentence length. These features offer complementary evidence with sentiment and moral signals.

F. Model Training Setup

We apply the classification using this expanded feature set which considers the moral, trajectory, emotion, and style features in combination with this. In our dataset we use X for feature matrix and y for the binary target variable. Stratified K-Fold Cross-Validation with $k = 5$ is utilized to make sure the classification of both classes is unbiased. This method ensures to balance class distributions in each split of data.

Additionally, `scale_pos_weight` (SPW) is utilized to take class imbalance into account which is a hyper-parameter, where:

$$SPW = \frac{N_{\text{negative}}}{N_{\text{positive}}}$$

N_{negative} and N_{positive} indicates number of non-manipulative and manipulative samples in the training fold. The class weighting balances minority class samples appropriately during the training.

G. Base Model and Hyperparameter Configuration

XGBoost [8][9] is used as the initial baseline model since it performs well on structured datasets due to its usage of gradient boosting along with the ability of regularization and sampling to control generalization. LightGBM [12][13] is introduced as another boosting model which has advantage of faster training speed and ability to process high dimension features due to its leaf-based growth and intrinsic class balancing.

Apart from that, more advanced optimisation and ensemble methods are utilised to optimise the models performance. Hyperparameters were tuned by Optuna with

Tree-structured Parzen Estimator (TPE). Also, we have utilised stacking ensemble by fitting diverse base learners such as XGBoost, Random Forest, and LightGBM, and using Logistic Regression as meta-learner to aggregate the probabilities of prediction [14][15] to achieve better result.

H. Feature Importants and Correlation Analysis.

To infer the importance of features proposed by this work, one must first select the best performing model through comparing it via cross-validation to identify and ultimately use the most interpretable models such as the Optuna-optimized XGBoost model [8][9]. The chosen model is then trained again using the full training data set to then perform a feature importance analysis to retrieve the weights. The top 25 features which contribute most are extracted and a visualization of features is presented that shows the dominant signal(s).

Finally, these features are split into 4 types of features for better interpretation: dynamic moral trajectory (the novel features), VADER sentiment, stylistic, eMFD static so we can determine the specific impact each has to the prediction the model is producing. Finally, the Spearman rank correlation is calculated to highlight the correlation coefficients (ρ) and p-values between individual features and our target value.

$$\rho_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between rank of feature value and rank of the target label for observation i and n is the total number of observations.

Integrating model-based importance rank with statistical correlation analyze in this methodology guarantees the extracted feature would be discriminative and have meaningful connection to emotion manipulation patterns.

IV. RESULT AND DISCUSSION

A. Feature Summary

Sentences were divided by the span “spacy” to all the more than 20 characters, the index for each document and sentence was attributed and was based on sentence-ordering within a document so as that analyses could be done on a trajectory-based. The total number of sentences per document were kept for computation. We ended up extracting a set of 28 features structured in two categories, in Table II.

TABLE II. DYNAMIC MORAL TRAJECTORY FEATURES GROUP

Group	Features
Document-Level Trajectory	avg_moral_density, peak_moral_density, tone_volatility, initial_hook_density, final_escalation, escalation_delta, switching_rate, moral_entropy, instability_index, vice_dominance_ratio, trajectory_slope, vice_virtue_ratio_volatility, peak_avg_ratio
Per-Foundation Statistics	virtue_mean, vice_mean, intensity_std $\times$ 5 foundations (care, loyalty, fairness, authority, sanctity)

B. Model Performance Comparison

To determine the best classification scheme, we train and tested four models based on 5-fold Stratified Cross Validation and evaluated on metrics namely: Accuracy, Macro-F1 and AUC-ROC. In table III the hyperparameters configurations are detailed.

TABLE III. HYPERPARAMETER CONFIGURATION

Parameter	XGBoost	LightGBM	XGBoost + Optuna
n_estimator	500	600	200-1000
max_depth	5	6	3-8
learning_rate	0.03	0.02	0.005-0.15
subsample	0.85	0.85	0.6-1.0
colsample_bytree	0.70	0.70	0.5-1.0
scale_pos_weight	N_neg/N_pos	N_neg/N_pos	N_neg/N_pos
Optimization	Manual	Manual	Optuna TPE 75 trials

The results are summarized in table IV.

TABLE IV. MODEL PERFORMANCE COMPARISON

Model	Accuracy	Macro-F1	AUC-ROC
XGBoost	0.735	0.733	0.820
LightGBM	0.737	0.734	0.819
XGBoost + Optuna	0.739	0.736	0.821
Stacking Ensemble	0.737	0.735	0.822

All the model performance performed very similar result. Stacking Ensemble performs little better than any of individual models by learning a complementary base-level ensemble. Stacking utilizes combination of base estimators to provide better prediction. Optuna optimized XGBoost presents almost the similar accuracy as the Stacking Ensemble which is a good score and very good among single estimators.

While LightGBM performs similarly with tuned XGBoost with leaf-wise growth strategy for high dimension data, the manually tuned XGBoost provides a very reasonable baseline value by confirming that dynamic moral trajectory feature works well. All the four approaches had very similar classification accuracy and these results indicates thatdynamic moral trajectory feature alone or in combo is sufficiently powerful to perform such classification.

C. Feature Important Analysis

The XGBoost model that optimized with Optuna, was then chosen because it's the best ‘one-handed model’ to interpret the feature. The top 25 most impactful features were identified and then further separated into four distinct feature categories: dynamic moral trajectory, VADER sentiment, stylistic, and static eMFD features.

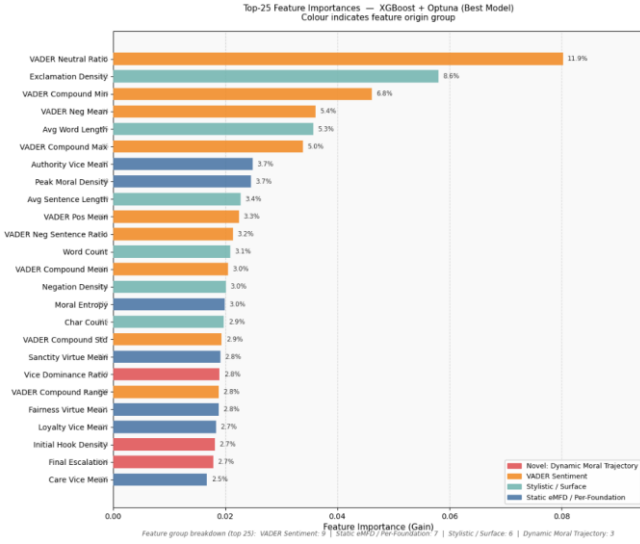


Fig. 2. Feature Importance

The results of feature importance indicated a clear dominance by VADER sentiment and stylistic features at the top. Specifically, VADER Neutral Ratio (9.9%) and Exclamation Density (9.1%) ranked as the two top predictive features. VADER sentiment featured 7 out of the top 25 features, while stylistic features represented 8 features and static eMFD per-foundation features captured 7 features in total. Dynamic moral trajectory features were comparatively the lowest contributors with Vice Dominance Ratio (2.9%), Final Escalation (2.8%), and Initial Hook Density (2.7%) appearing amongst the top 25 (3 features in total).

Dynamic moral trajectory features did not significantly outrank features in other categories, yet their contribution among the top 25 features along with moral entropy (3.0%), demonstrated how intra-document moral patterns offered an additional source of discriminative information. Furthermore, the substantial impact of VADER sentiment and stylistic features, including patterns like neutral masking, amplifying punctuation and negating constructs, could infer that superficial rhetorical phenomena is an extremely important factor to detecting manipulation.

D. Spearman Correlation Analysis

In order to assess the statistical relationship of engineered features with manipulation label, we used spearman rank correlation over the selected features across all the groups (Table V).

Feature	Group	ρ	p-value
Tone Volatility	Dynamic Moral Trajectory	0.1713	0.000000
Instability Index	Dynamic Moral Trajectory	0.1710	0.000000
Vice/Virtue Ratio Volatility	Dynamic Moral Trajectory	0.1696	0.000000

Switching Rate	Dynamic Moral Trajectory	0.1593	0.000000
Final Escalation	Dynamic Moral Trajectory	0.1125	0.000000
Initial Hook Density	Dynamic Moral Trajectory	0.1116	0.000000
Trajectory Slope	Dynamic Moral Trajectory	0.0212	0.000330
Escalation Delta	Dynamic Moral Trajectory	0.0101	0.088382
Avg Word Length	Stylistic / Surface	-0.2219	0.000000
VADER Pos Sentence Ratio	VADER Sentiment	0.0051	0.391374

Dynamic moral trajectory features have the most significant and consistent correlations with the manipulation label. Tone Volatility ($\rho = 0.1713$), Instability Index ($\rho = 0.1710$), Vice/Virtue Ratio Volatility ($\rho = 0.1696$) and Switching Rate ($\rho = 0.1593$) show significant positive correlation ($p < 0.001$) supporting that the emotional manipulation style is accompanied by instability and dynamism in moral framing of the text.

The highest overall correlation for a single feature is with Avg Word Length ($\rho = -0.2219$) suggesting that manipulate text has simpler word construction with less length. It should be mentioned that, there are some other features like Escalation Delta ($\rho = 0.0101$, $p = 0.088$) and VADER Pos Sentence Ratio ($\rho = 0.0051$, $p = 0.391$) which are not found to have a linear association with label, however these have contributed to model's performance. In conclusion, the spearman's rank correlation confirms the statistical viability of our proposed dynamic moral trajectory features for detecting the patterns in text related to emotional manipulation.

E. Discussion

The experimental results provide evidence that our Dynamic Moral Trajectory Engineering method captures the intra-document moral dynamics in the form of a number of measurable and statistically significant features. Spearman correlation demonstrates that trajectory features, such as Tone Volatility ($\rho = 0.1713$), Instability Index ($\rho = 0.1710$), Vice/Virtue Ratio Volatility ($\rho = 0.1696$) and Switching Rate ($\rho = 0.1593$), continue to hold positions among the most correlated features with the manipulation label, thus validate the hypothesis that texts are deliberately escalated by shifting moral frames across sentences in the form of fluctuations over sentences. Our dynamic moral trajectory features bring about a significant leap forward than static document-level, as the features capture moral dynamics instead of the static average values over the document.

Feature importance shows that VADER sentiment and stylistic features yield the most influential factors to predict label, but this is justified given the datasets used which Twitter and propaganda texts contain sufficient surface-level rhetorical features. In the case where surface features are carefully concealed in subtle manipulation scenarios, our

dynamic moral trajectory features are expected to yield even better discrimination performance.

Several issues should be pointed out. The binary label design is based on MFT theory, not annotations from human experts; there by, label noises exist. Its adaptability on non-twitter and non-propaganda texts is unknown. Extensions on other datasets and different working scenarios such as multilingual and real-time detection settings in the future would better verify the effectiveness of Dynamic Moral Trajectory Engineering.

V. CONCLUSION

We presented the Moral Hijacking Detector, an NLP framework to detect emotionally manipulative text based on Moral Foundations Theory, dynamic feature engineering, and ensemble learning. Our work proposed the Dynamic Moral Trajectory Engineering that captures dynamic patterns of how moral framing is applied in sequence through documents, including tone volatility, switching rate, instability index, and escalation as a document evolves, which is unable to be addressed by static moral scoring.

In empirical experiments, the single model XGBoost with Optuna performs best in classification with all tested configurations. Spearman correlation analysis confirms that dynamic moral features such as Tone Volatility ($\rho = 0.1713$), Instability Index ($\rho = 0.1710$) and Switching Rate ($\rho = 0.1593$) are significantly relevant to manipulations compared to just surface-level features.

The major innovation of this study is moving the detection of emotional manipulation from static analysis of morality to dynamic and explainable, theory-informed detection. These novel trajectory features offer an unprecedented way to examine the dynamics and incremental nature with which a writer alters moral stance over time and provide both new methods and new understandings about emotional manipulation.

This approach has practical applications for use in explainable AI for education applications, content moderation tools and misinformation detection engines. Potential study weaknesses involve the use of labels derived from a theory instead of directly from humans, and limitation to analysis of propaganda texts and twitter data that may not transfer readily to other genres and languages.

For future work, one could compile large fine-grained human annotated datasets, explore more broadly in multilingual settings and use deep learning models in the proposed framework in order to capture sequential dynamics more successfully.

REFERENCES

- [1] Dawson, K. J., Han, H., & Choi, Y. E. R. (2023). How are moral foundations associated with empathic traits and moral identity? *Current Psychology*, 42(13), 10836–10848. doi: 10.1007/s12144-02102372-5
- [2] Goenka, S., & Thomas, M. (2024). Moral foundations theory and consumer behavior. *Journal of Consumer Psychology*, 34(3), 536–540. doi: 10.1002/jcpy.1429
- [3] Maninger, T., & Shank, D. B. (2022). Perceptions of violations by artificial and human actors across moral foundations. *Computers in Human Behavior Reports*, 5. doi: 10.1016/j.chbr.2021.100154
- [4] Culiberg, B., Cho, H., Kos Koklic, M., & Zabkar, V. (2023). The Role of Moral Foundations, Anticipated Guilt and Personal Responsibility in Predicting Anti-consumption for Environmental Reasons. *Journal of Business Ethics*, 182(2), 465–481. doi: 10.1007/s10551-021-05016-7
- [5] Dogruyol, B., Velioglu, İ., Bayrak, F., Acem, E., Isler, O., & Yilmaz, O. (2024). Validation of the moral foundations questionnaire-2 in the Turkish context: exploring its relationship with moral behavior. *Current Psychology*, 43(29), 24438–24452. doi: 10.1007/s12144-024-06097-z
- [6] Telkamp, J. B., & Anderson, M. H. (2022). The Implications of Diverse Human Moral Foundations for Assessing the Ethicality of Artificial Intelligence. *Journal of Business Ethics*, 178(4), 961–976. doi: 10.1007/s10551-022-05057-6
- [7] Banker, S., Park, J., & Chan, E. Y. (2023). The moral foundations of cryptocurrency: evidence from Twitter and survey research. *Frontiers in Psychology*, 14. doi: 10.3389/fpsyg.2023.1128575
- [8] Hakkal, S., & Lahcen, A. A. (2024). XGBoost To Enhance Learner Performance Prediction. *Computers and Education: Artificial Intelligence*, 7. doi: 10.1016/j.caeai.2024.100254
- [9] Zhang, P., Jia, Y., & Shang, Y. (2022). Research and application of XGBoost in imbalanced data. *International Journal of Distributed Sensor Networks*, 18(6). doi: 10.1177/15501329221106935
- [10] Hopp, F. R., Fisher, J. T., Cornell, D., Huskey, R., & Weber, R. (2021). The extended Moral Foundations Dictionary (eMFD): Development and applications of a crowd-sourced approach to extracting moral intuitions from text. *Behavior Research Methods*, 53(1), 232–246. doi: 10.3758/s13428-020-01433-0
- [11] Zhang, Y., Chen, L., & Xie, M. (2025). How do NPOs' topics and moral foundations in gun-related issues influence public engagement on Twitter? *Internet Research*, 35(3), 981–1001. doi: 10.1108/INTR-11-2022-0909
- [12] Rufo, D., Debelee, T. G., Ibenhal, A., & Negera, W. G. (2021). Diagnosis of diabetes mellitus using gradient boosting machine (LightGBM). *Diagnostics*, 11(9), Article 1714. doi: 10.3390/diagnostics11091714
- [13] Talebi, H., Bardsiri, A. K., & Bardsiri, V. K. (2025). Developing a hybrid machine learning model for employee turnover prediction: Integrating LightGBM and genetic algorithms. *Journal of Open Innovation: Technology, Market, and Complexity*, 11(2), Article 100557. doi: 10.1016/j.joitmc.2025.100557
- [14] Yoon, T., & Kang, D. (2023). Multi-modal stacking ensemble for the diagnosis of cardiovascular diseases. *Journal of Personalized Medicine*, 13(2). doi: 10.3390/jpm13020373
- [15] Gurcan, F. (2025). Enhancing breast cancer prediction through stacking ensemble and deep learning integration. *PeerJ Computer Science*, 11. doi: 10.7717/peerj-cs.2461
- [16] Isnan, M., Elwirehardja, G. N., & Pardamean, B. (2023). Sentiment analysis for TikTok review using VADER sentiment and SVM model. *Procedia Computer Science*, 227, 168–175. Elsevier B.V. doi: 10.1016/j.procs.2023.10.514
- [17] Saoualili, A., Safaa, L., Bouhatous, A., Bidan, M., Perkumienė, D., Aleinikovas, M., & Perkumas, A. (2024). Exploring the tourist experience of the Majorelle Garden using VADER-based sentiment analysis and the latent Dirichlet allocation algorithm: The case of TripAdvisor reviews. *Sustainability (Switzerland)*, 16(15). doi: 10.3390/su16156378